Matched-Filter Recovery of Pasted Noise and a Twelve-Member Sound-Event Ensemble for Indic Speech Noise Event Detection

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Datathon@IndoML 2026, Track 1 (Noise Event Detection)
Code, heads and submissions: https://github.com/syehsyoh-dokling/NoiseDetection

***Abstract*—Track 1 asks for the onset and offset of every noise event in Vaani field recordings, scored by an event-based F1 (both boundaries within 20 % of the reference duration, at least 50 ms) plus a per-clip segment Dice. Our final submission (Codabench submission 994586) placed 5th of 82 teams with a combined score of 1.600895 (F1 0.780264, Dice 0.820631; maximum 2.0). The synthetic clips appear to be clean speech with noise samples pasted from a finite library; the validation release ships the clean references, so mixture minus reference recovers the pasted waveforms, and a normalised cross-correlation (NCC) matched filter over 1,198 such templates locates them in the test clips. Natural clips are handled by a frame-level sound-event detector (SED): a weighted mean of twelve members, each a bidirectional GRU (BiGRU) head over a frozen or fine-tuned Whisper, AST, BEATs or Dasheng encoder, decoded with per-event peak-relative boundaries. We report the leaderboard lineage, out-of-fold component scores (fold spread, per-category recall), compute cost, and the reproduction after the fine-tuned checkpoints were lost.**

I. Training Data and Preprocessing

Vaani data. The released validation set holds 2,501 clips (3.51 h, 16 kHz mono): 1,806 natural field recordings and 695 synthetic clips, each synthetic clip with its clean reference, from 28 states, 141 state/district pairs and 27 languages, with 5,157 labelled events in seven top-level categories (human non-speech 1,924; vehicle/traffic 1,281; animal 1,142; baby/child 496; phone/signal/alarm 188; singing/music 68; appliance/machine 58). The test set holds 5,517 unlabelled clips; no test labels were released or accessed.

External data (permitted by the rules [1], which allow additional public or proprietary data). We used the Gold tier of the Vaani noise-event dataset [2], [3] (10,772 clips, 19,414 verified-timestamp events; CC BY 4.0, gated access which we hold) for every member except wsA and wlP, and its Silver tier (unverified timestamps) for wlHS (14,529 clips from the Hugging Face release) and bFS (the first 12,000 clips of the Kaggle copy ayanokoji2028/vaani-noise-hf of [2], HF file names), at loss weight 0.4 (member tags: Table I). Overlap was checked by Vaani file stem, a unique clip id: Gold shares no clip with the test set (0 of 5,517) or the validation set (0 of 2,501) and no recording session (file-name prefix) with any test clip; 13 Gold clips share a session with a validation clip. Both Silver extractions exclude every validation and test file name. Speaker/district overlap with the test set was not checked (domain, not label, leakage).

Test-side information. Two uses are disclosed. (a) The leaderboard score of each upload was the final selection criterion (Sec. III); Codabench [1] lists 47 uploads by our team (46 scored; 995069, 17 Oct, after the final, failed) and our first upload (921278, 10 Sep) is no longer listed, so at least 48 were made (quota 5 per day, 100 in total). (b) Member wlP was trained on pseudo-labels [4] that three validation-only Whisper heads (small, medium, large, weighted 1/1/2) produced on the 3,353 test clips without a library hit, a transductive use of unlabelled test audio (the organisers list semi-supervised use of unlabelled Vaani audio as a novelty direction, which we read as covering these clips). The fold-f student only sees labels from teachers that never saw fold f (leak-free OOF); Table III reports the ensemble without it.

Template library. Outside the pasted spans ($\pm$150 ms) the residual (mixture minus clean reference) of a synthetic validation clip is zero at most samples (per-clip median 0, maximum $4.4{\times}10^{-3}$ at full scale 1), so every labelled synthetic event cut from the residual is the pasted noise sample. Near-duplicates (NCC > 0.995) were merged, leaving 1,198 templates, each with its source clips for leave-one-clip-out (LOO) evaluation.

Features. Every detector works on a 20 ms frame grid. For each encoder, the tokens of three or four selected layers are averaged over frequency (AST, BEATs, Dasheng), layer-normalised per frame, averaged over layers and interpolated from their centres onto the grid; Whisper's 20 ms frames are used directly. Targets are eight binary channels per frame (any noise plus the seven categories): 24 % of natural reference events overlap in time with an event of another category, which the event-F1 term scores separately.

Validation protocol. All model selection used 5-fold cross-validation grouped by state/district (GroupKFold; 500–501 clips, 27–29 districts per fold), to mimic a test set from unseen districts; every validation number below is out-of-fold (OOF) with the test decoding, the library in LOO mode and the null levels cross-fitted (two halves of the natural clips calibrate each other; synthetic clips use all natural clips). Fusion weights and decode parameters were fitted on these same OOF predictions (no nested CV), so the OOF scores are optimistic for those hyperparameters.

TABLE I

THE TWELVE MEMBERS. HEAD: BIGRU (HIDDEN 192, DROPOUT 0.3) + LINEAR TO 8 CHANNELS; BINARY CROSS-ENTROPY (MASKED TO THE CLIP'S FRAMES, PER-CHANNEL WEIGHTS AS IN THE CODE), SPECAUGMENT-STYLE MASKING [15], ADAMW [16] (WD 10^{-2}), ONE-CYCLE PEAK 10^{-3}, LAST EPOCH KEPT, 5 DISTRICT FOLDS (TEST = MEAN OF THE 5 FOLD HEADS). V VALIDATION FOLDS, G GOLD, S SILVER (W 0.4), P PSEUDO-LABELS (W 0.5). FT = ENCODER FINE-TUNED (LR 2×10^{-5}, HEAD 10^{-3}, FP16 AUTOCAST, GRADIENT CHECKPOINTING).

Tag	Encoder	Layers	Data	Ep. / batch / crop	w
wsA	whisper-small, frozen	3,6,9,12	V	25 / 16 / 600 fr	0.5
wlP	whisper-large-v3, frozen	8,16,24,32	V+P	25 / 16 / 600 fr	3
wlH/wlH2/wlH3	whisper-large-v3, frozen	8,16,24,32	V+G	8 / 8 / 400 fr (seed 0/1/2)	2/6/1
wlHS	whisper-large-v3, frozen	8,16,24,32	V+G+S	6 / 8 / 400 fr	6
aH, bH, dH	AST, BEATs, Dasheng, frozen	4,8,12	V+G	8 / 8 / 400 fr	8, 8, 4
aFG	AST, all 12 blocks ft	4,8,12	V+G	3 / 8 / 10.24 s	8
bFG	BEATs ft, patch embed frozen	4,8,12	V+G	3 / 8 / 10.24 s	8
bFS	BEATs ft, no hold-out	4,8,12	all V+G+S	2 / 8 / 10.24 s	4

II. PROPOSED METHOD

Background. Frame-level CRNN classifiers [5], [6] are a standard approach to SED; Whisper's encoder keeps background-sound identity [7], and AudioSet [8] transformers (AST [9], BEATs [10], Dasheng [11]) are strong audio encoders. The synthetic half is detection of a known signal, for which the matched filter is SNR-optimal in white Gaussian noise [13]; we use its normalised form, fast NCC [14].

Matched filter. For a test clip x and template T_i, the matcher computes the NCC $c_i(s)$ at every sample lag s with FFTs on the GPU. A per-template null level ν_i is the 0.999 quantile of $\max_s c_i(s)$ over the natural validation clips. Candidates are kept when the margin $c_i(s) - \nu_i \geq 0.1$ and selected greedily by margin with at most 30 % mutual overlap. In LOO mode on validation the matcher alone scores F1 0.9819 / Dice 0.9723 on the 695 synthetic clips (Table III); it fires on 687 of them and on 73 of the 1,806 natural clips.

SED ensemble. Table I lists the members. The ensemble probability is the weighted mean

$$p = \Big(\sum\nolimits_m w_m p_m\Big) \Big/ \sum\nolimits_m w_m. \tag{1}$$

The weights of the eleven members with OOF predictions were set by coordinate descent over a coarse grid on the natural clips (`tune_weights.py`; the matcher dominates the synthetic ones); bFS's weight was a leaderboard probe. Members were added one at a time; our working rule was that a candidate had to raise the pooled OOF score with at least four of five district folds up before it was given one upload ("one-slot probe"), which decided; the per-member OOF records were not retained, so this is stated as practice, not demonstrated.

Decoding and fusion. Natural clips use two branches: $p_h = \min(p_{\text{hns}}, p_0)$ for human non-speech (hns) and $p_o = \min(\max_{c\neq\text{hns}} p_c, p_0)$ for the other categories, p_0 being the any-noise channel. Each branch is decoded peak-relatively: a region is detected where the median-filtered curve (window 15 / 7 frames) clears a floor of 0.2, is kept if its peak reaches 0.34 / 0.30, and both boundaries are placed where the curve crosses α times that event's own peak ($\alpha = 0.6$ / 0.5); introduced before upload 924220, this was worth +0.029 natural OOF event F1 over one global threshold on the same probabilities. Events shorter than 0.1 s are dropped, gaps below 0.1 s / 2.0 s are closed, and the branches are merged with duplicate removal at IoU 0.5. A clip with any library hit is treated as synthetic: its events are the template events plus any SED event (any-noise channel, threshold 0.7, median 3, minimum 0.1 s, gap 0.04 s) that does not overlap them (`models/fusion_Q1_bFSadd4.json`). The system is offline and non-causal: the heads are bidirectional, the decoder needs the whole event, the encoders see the whole clip.

TABLE II

LEADERBOARD (LB) LINEAGE: 10 OF OUR AT LEAST 48 UPLOADS. OOF: OUT-OF-FOLD SCORE (`RUN.PY TUNE`: LOO LIBRARY, CROSS-FITTED NULLS); ΔOOF: GAIN OVER THE PREVIOUS ROW (FOLDS UP OF 5), WHERE RECORDED. MED / THR: MEDIAN WINDOW (FRAMES) / THRESHOLD OF THE HUMAN-NON-SPEECH BRANCH. Q1 = FINAL CONFIGURATION WITHOUT BFS.

Upload	Change	OOF	ΔOOF	LB F1	LB Dice	LB comb.
921278[c]	first upload, 10 Sep	–	–	0.7113	0.7478	1.4591
924220	v14: 6 Whisper + 2 AST frozen members	1.4466	–	0.7752	0.8055	1.5808
929944	+ aFG:8, aA dropped[b]	–	–	0.7739	0.8130	1.5868
932046	+ bH:8	–	–	0.7741	0.8148	1.5890
932698	+ bFG:8, dH:4	–	–	0.7752	0.8187	1.5938
933670	boundary calibr. (med 11, thr 0.32)	–	+0.0024 (5)	0.7778	0.8194	1.5972
936587	med 13, thr 0.34	–	+0.0017 (4)	0.7799	0.8201	1.5999
991824	med 15	–	–	0.7802	0.8198	1.6000
991838	other-branch gap 2.0 (Q1)	1.4787	+0.0007 (4)	0.7806	0.8195	1.6002
994586	**+ bFS:4 (final)**	n/a[a]	–	**0.7803**	**0.8206**	**1.6009**

[a] bFS trains on all validation clips: no OOF estimate; weight set by a one-slot leaderboard probe. [b] aA, the second AST member of v14, dropped after its artifacts were lost (v14 without it: 1.4474 OOF). [c] No longer listed on Codabench; from our records. Differences below 0.001 between the last rows are smaller than the leaderboard noise we did not quantify.

Compute and software. Python 3.13.6, PyTorch 2.14.0 (CUDA 12.6). Frozen heads (on cached float16 features), all aFG folds and bFG folds 0–2 were trained on one RTX 4060 Laptop GPU (8 GB); bFG folds 3–4 and bFS on Kaggle T4 GPUs. On the T4 (retrain logs): one aFG fold 64–67 min, one bFG fold about 55 min, bFS 75 min; one fine-tuned encoder pass over the 5,517 test clips 4.5–5 min (about 55 min for the eleven); the five frozen-backbone passes plus downloads under 1.4 h; the matched filter 84.6 min on the laptop GPU (FFT up to 2^{19}). Pretrained encoders: openai/whisper-small and whisper-large-v3 [17] (Apache 2.0), MIT/ast-finetuned-audioset-10-10-0.4593 [9] (BSD 3), BEATs iter3+ AS2M [10] (MIT), mispeech/dasheng-base [11] (Apache 2.0); all public. The original encoders are not redistributed; the eleven fine-tuned AST/BEATs checkpoints are released as derivatives under BSD 3 / MIT with the original notices.

III. RESULTS AND CONCLUSION

Leaderboard. Table II traces the system from the first upload (1.4591) to the final 1.6009 (our highest listed upload). The changes up to v14 (more members, Gold data, the split and peak-relative decodes) are not separable in our records and give most of the gain (to 1.5808); fine-tuning AST, dropping aA and adding BEATs and Dasheng add +0.0131; OOF-tuned decoding changes +0.0063; bFS +0.0007, below what we can distinguish from noise. The gap to rank 1 (1.6173) is 0.0164:

TABLE III
OOF COMPONENT SCORES OF Q1 ON VALIDATION: COMBINED (F1 / DICE) ON ALL 2,501, THE 1,806 NATURAL AND THE 695 SYNTHETIC CLIPS; FOLDS: MEAN ± SD OVER THE 5 DISTRICT FOLDS (RANGES 1.379–1.590, 1.378–1.583, 1.169–1.243). LOO LIBRARY, CROSS-FITTED NULLS.

System	All	Natural	Synthetic	Folds
Q1 full	1.4787 (.6894/.7894)	1.1570 (.4465/.7105)	1.9868 (.9925/.9943)	1.476±.077
without wlP	1.4759 (.6872/.7887)	1.1527 (.4431/.7096)	1.9866 (.9923/.9943)	1.474±.075
retrained aFG/bFG	1.4771 (.6886/.7885)	1.1542 (.4450/.7092)	1.9869 (.9925/.9944)	–
SED only	1.2005 (.4500/.7505)	1.1409 (.4331/.7077)	1.3378 (.4761/.8618)	1.201±.031
Templates only	0.8871 (.5976/.2895)	0.0691 (.0424/.0268)	1.9542 (.9819/.9723)	–

TABLE IV
NATURAL CLIPS, Q1 OOF: RECALL OF THE 2,944 REFERENCE EVENTS PER CATEGORY UNDER THE METRIC'S BOUNDARY TOLERANCE (PREDICTIONS CARRY NO CATEGORY); n, MEDIAN REFERENCE DURATION.

	human	vehicle	animal	baby	alarm	music	machine	all
n	1,254	655	594	265	110	40	26	2,944
recall	0.622	0.282	0.269	0.215	0.300	0.225	0.231	0.418
med. dur. (s)	0.32	0.70	0.79	0.70	0.60	2.78	6.23	–

F1 0.0092, Dice 0.0072. Selection was leaderboard-guided, so the final score carries an unquantified optimistic bias. OOF and LB scores of one configuration differ by 0.12 (1.4787 vs 1.6002); we attribute this to the test set being 39 % synthetic-like (2,164 of 5,517 clips with a library hit, against 28 % truly synthetic and 30 % with a LOO hit in validation) and to the test run using the full library.

Components and errors. Table III: on synthetic clips the matcher alone reaches 1.9542 and the SED alone 1.3378; on natural clips the SED alone gives 1.1409, the matcher alone 0.0691, and the fusion adds +0.016 (template events in the 73 natural clips with a hit). wlP is worth +0.0028 pooled; the fold spread (sd 0.077) is far larger, but the paired difference is positive on 4 of 5 folds (+0.0066 to −0.0008). Table IV shows where the natural-clip F1 of 0.45 is lost: 62 % of human non-speech events (short, median 0.32 s) are recovered but only 22–30 % of vehicle, animal, baby, alarm, music and machine events, which are longer and must meet the 20 % tolerance at both ends (on a pre-v14 ensemble only 12.9 % of natural reference events were truly missed, 52 % failed a boundary tolerance). Also tried and absent from the final system: a soft-Dice loss term (wlHSD, refuted out-of-fold), a stacker, CLAP and CED [12] members, a fine-tuned Whisper (wFG), a fine-tuned Dasheng (dFG) and a second AST seed (configurations in `models/fusion_*.json`; the basis of each rejection was not uniformly recorded).

Reproduction. After the final upload the workspace lost the backbone weights, the fine-tuned checkpoints, all cached test probabilities and the test template candidates; the 45 frozen fold heads, configurations, template library and scored zip survived. We retrained aFG, bFG and bFS with the identical scripts and their fixed seeds on Kaggle T4 GPUs (aFG and bFG folds 0–2 were originally trained on the laptop GPU), re-ran the matcher, and recomputed the frozen members from the released heads and re-downloaded public encoders. The retrained aFG/bFG give 1.4771 OOF against 1.4787 (Table III); the rebuilt submission flags the same 2,164 clips as synthetic-like, is identical on 41.8 % of the clips (2,008 of 2,164 synthetic-like, 298 of 3,353 natural-like; 13,442 vs 13,451 events) and scores 1.9858 (F1 0.9943 / Dice 0.9915) under the metric with the scored file as reference. The released fine-tuned checkpoints (eleven files, 3.96 GB, release assets) are thus not bit-identical to the scored ones (different GPU, nondeterministic kernels); bFS has no OOF estimate, so its equivalence rests on that agreement. The pseudo-label file and teacher heads of wlP are not released.

Limitations. The matched filter relies on the test set reusing the validation noise library; with fresh noise waveforms the SED branch alone (1.2005 OOF) would carry the score. The ensemble (five frozen backbones, eleven fine-tuned encoders) was chosen for score, not cost, and is offline and non-causal; no real-time variant was built or measured. wlP uses test audio transductively. Natural-clip event F1 (0.45 OOF, recall 0.22–0.30 for non-human categories) is the weakest part.

Conclusion. Known-signal detection on the synthetic half and frame-level SED with per-event boundary placement on the natural half gave a 5th-place system, fully released.

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